

## Program – 8: Create Kubernetes Pods using YAML Descriptors

### Description

This program demonstrates deploying a Node.js application inside a Kubernetes Pod using declarative YAML instead of imperative commands.

### Procedure

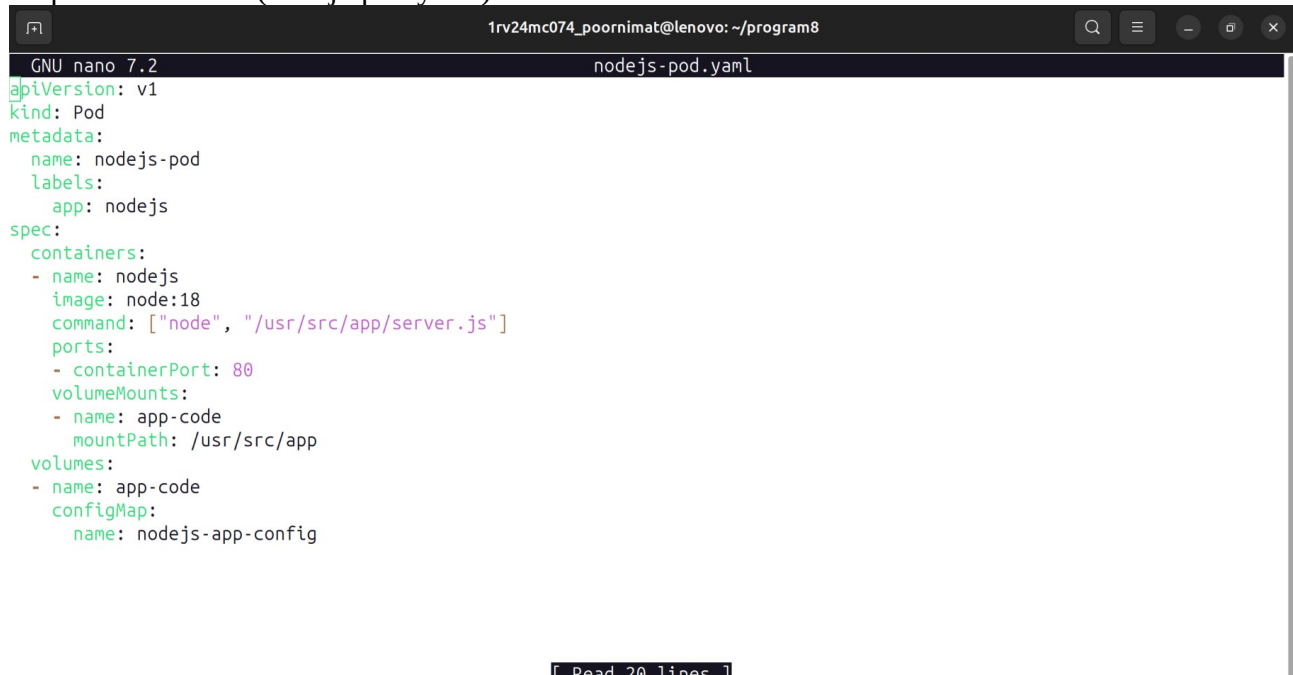
#### Step 1: Create Node.js App (server.js)

```
const http = require("http");
const port = 3000;
const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});
server.listen(port);
```

#### Step 2: Create ConfigMap (nodejs-configmap.yaml)

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
const http = require("http");
const port = 3000;
const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});
server.listen(port);
```

#### Step 3: Create Pod (nodejs-pod.yaml)

A screenshot of a terminal window with a dark background. The window title is "1rv24mc074\_poornimat@lenovo: ~/program8". The terminal shows the contents of a file named "nodejs-pod.yaml" being edited with "GNU nano 7.2". The YAML content is as follows:

```
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      ports:
        - containerPort: 80
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

At the bottom of the terminal, a status bar indicates "Read 20 lines |".

#### Step 4: Apply Files

```
kubectl apply -f nodejs-configmap.yaml
```

```
kubectl apply -f nodejs-pod.yaml
```

#### Step 5: Port Forward

```
kubectl port-forward pod/nodejs-pod 3000:3000
```

Open: <http://localhost:3000>

#### Expected Output

Hello from Node.js running inside Kubernetes!

```
1rv24mc074_poornimat@lenovo:~/program8$ kubectl apply -f nodejs-cofigmap.yaml
configmap/nodejs-app-config configured
1rv24mc074_poornimat@lenovo:~/program8$ kubectl apply -f nodejs-p
error: the path "nodejs-p" does not exist
1rv24mc074_poornimat@lenovo:~/program8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
1rv24mc074_poornimat@lenovo:~/program8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
]
```

